# Module Q03 — Is Reasoning Real, a Property of Language, a Reflection, or an Illusion?

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DLS 2026 Teaching Staff

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# Framing the Question

The student asks four distinct hypotheses, often conflated:

- **Real**: the model executes genuine inference steps that cause its answer.
- **Property of language**: "reasoning" is latent in the statistics of text; the model inherits it.
- **Reflection**: the model mirrors reasoning patterns present in training data without owning the process.
- **Illusion**: the appearance of reasoning is a post-hoc narrative, not the mechanism.

Claim of this module: these are not mutually exclusive, and the empirical evidence partitions across them depending on *task*, *scale*, and *what we measure*.

# Evidence That Something Real Emerges

- Chain-of-thought (CoT) prompting materially improves multi-step arithmetic and symbolic tasks, with gains sharpest at scale [1].
- Self-consistency (sampling multiple chains, taking the majority) further lifts accuracy, suggesting the chains carry task-relevant computation [2].
- Scratchpads let transformers perform long multi-step computations by externalizing intermediate state [3].

**Interpretation**: intermediate tokens expand the effective serial computation budget — a mechanistic reason CoT helps rather than mere decoration.

# Evidence That It Is a Property of Language / Statistics

- Many "reasoning" gains track surface features: answer frequency, term overlap, and token-level priors.
- Benchmark performance can shift sharply with superficial perturbations, implying reliance on distributional cues rather than abstract rules [5].
- Capabilities ride on scale and data coverage, consistent with reasoning being recoverable *structure* in language rather than a separate faculty [4].

So some of what looks like reasoning is competence *compressed into* the language distribution.

# Evidence That It Is a Reflection or an Illusion

- CoT explanations can be *unfaithful*: models reach the same answer even when the stated reasoning is corrupted, and are influenced by biasing features they never mention [6].
- Verbalized reasoning need not be the causal path to the output — the narrative may be reconstructed after the computation [6].
- Robustness gaps on perturbed inputs suggest pattern-reflection over principled inference [5].

Caution: "the explanation is unfaithful" refutes *transparency*, not necessarily the existence of internal computation.

# Disentangling: Mechanism vs. Explanation

Two separable claims:

1. **Does internal computation occur?** Interpretability work finds structured, reusable circuits (e.g. induction heads implementing in-context copying/completion), a concrete mechanism underlying in-context generalization [7].
2. **Is the stated chain the mechanism?** Often not — faithfulness is not guaranteed [6].

$$\text{observed reasoning} = \underbrace{f_{\text{circuit}}(x)}_{\text{real computation}} + \underbrace{g_{\text{verbalize}}(x)}_{\text{possibly post-hoc}}$$

Measuring only text conflates these two terms.

# A Working Synthesis

| Hypothesis | True for... | Key evidence |
|---|---|---|
| Real | serial multi-step tasks | [1, 3] |
| Property of language | broad competence | [4] |
| Reflection | perturbation-fragile tasks | [5] |
| Illusion (of transparency) | verbal CoT fidelity | [6] |

**Bottom line**: real *computation* happens (circuits, extended serial depth), it is largely a *property learned from language*, and the *verbal explanation* is partly illusory. The illusion is about *introspective reporting*, not about the absence of mechanism.

# How to Test the Claim Yourself

- **Faithfulness probes**: perturb or bias the input; check whether the answer changes while the stated chain does not [6].
- **Ablate the scratchpad**: remove intermediate tokens and measure the accuracy drop to estimate the real serial-compute contribution [3].
- **Self-consistency spread**: variance across sampled chains diagnoses whether computation or priors dominate [2].
- **Mechanistic probing**: look for circuits (e.g. induction heads) correlated with the capability [7].

# Takeaways

- The question is under-specified until you fix *what* you measure.
- Internal multi-step computation is real and mechanistically visible [7]; it is learned from the distribution of language [4].
- The emitted reasoning trace is a useful but *unreliable* window into that computation [6].
- Design evaluations that separate *mechanism* from *explanation* rather than scoring narratives.

Reasoning is real as computation, linguistic in origin, and partly illusory as self-report.

# References


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